

Artificial Intelligence

AI2002

Lecture 8

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FAST NUCES CFD

Simulated Annealing

- Idea:
 - Escape local maxima by allowing some “bad” moves but gradually decrease the size and frequency of the bad moves
 - In thermodynamics, the probability to go from a state with energy E_1 to a state with energy E_2 is given by

$$p = e^{\frac{(E_2 - E_1)}{kT}} = e^{\frac{-(E_1 - E_2)}{kT}}$$

Simulated Annealing

- Where
- E is Euler number
- T is a “temperature” controlling the probability of downwards steps
- K is Boltzmann’s constant
 - Relating energy and temperature; with appropriate choice of units, it will be equal to 1.

Simulated Annealing

- Let E represent the objective function value (also known as energy).
- If $\Delta E = E_{next} - E_{current} > 0$ probability of accepting state with a better object function. (always go upward)
- If $\Delta E = E_{next} - E_{current} < 0$ probability of accepting state with a worse object function is.

$$P(\text{Accept Next}) = e^{\frac{\Delta E}{T}}$$

T = temperature at that time.

Simulated Annealing

function **SIMULATED-ANNEALING**(*problem*, *schedule*) **returns** a solution state

inputs: *problem*, a problem

schedule, a mapping from time to “temperature”

local variables: *current*, a node

next, a node

T, a “temperature” controlling prob. of downward steps

current $\leftarrow$ MAKE-NODE(INITIAL-STATE[*problem*])

for *t* $\leftarrow$ 1 **to** ∞ **do**

T $\leftarrow$ *schedule*[*t*]

if *T* = 0 **then** **return** *current*

next $\leftarrow$ a randomly selected successor of *current*

$\Delta E \leftarrow$ VALUE[*next*] - VALUE[*current*]

if $\Delta E > 0$ **then** *current* $\leftarrow$ *next*

else *current* $\leftarrow$ *next* only with probability $e^{\Delta E/T}$

Similar to hill climbing,
but a random move instead
of best move

case of improvement, make
the move

Otherwise, choose the move with probability that
decreases exponentially with the “badness” of the move.

Simulated Annealing

- The idea is that probability decreases exponentially with $E_2 - E_1$ increasing,
- The probability gets lower as temperature decreases
- If the ***schedule*** lowers T slowly enough, the algorithm will find a global optimum with probability approaching 1.

Example:

- Consider there are 3 moves available, with changes in the objective function of

$$\Delta E_1 = -0.1, \Delta E_2 = 0.5, \Delta E_3 = -3$$

- Pick a move randomly:
 - if ΔE_2 is picked, move there. $e^{-\frac{\Delta E}{T}}$
 - if ΔE_1 or ΔE_3 are picked, probability of move =
 - move 1: $\text{prob1} = e^{-0.1} = 0.9$,
 - i.e., 90% of the time we will accept this move
 - move 3: $\text{prob3} = e^{-3} = 0.0497$
 - i.e., 5% of the time we will accept this move

Role of Temperature

- High Temperature
 - Allows the algorithm to explore the solution space freely
- Low Temperature
 - Narrows down the exploration, focusing on refining solutions

Simulated Annealing

- T = “Temperature” Parameter
- **If T is high** $\Rightarrow$ the probability of “locally bad” move is higher
- **If T is low** $\Rightarrow$ the probability of “locally bad” move is lower
- typically, T is decreased as the algorithm runs longer
 - i.e., there is a “temperature schedule”

Question

Simulated annealing will find a global optimum with probability approaching 1 if you lower T slowly enough. Please explain what simulated annealing will do in each of the following case.

1. What happens if you lower T very quickly?
2. What happens if you never Lower T ?

Question

Simulated annealing will find a global optimum with probability approaching 1 if you lower T slowly enough. Please explain what simulated annealing will do in each of the following case.

1. What happens if you lower T very quickly? [Low Temperature]

You will very likely stuck in the local maximum.

2. What happens if you never Lower T ? [High Temperature]

You will always have a non- zero probability of going downhill. Therefore, even if you reach the global maximum you won't stay there.

Question

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1. What happens if you lower T very quickly?
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Example:

- If Simulated Annealing perform if the temperature, T , is always very close to 0 then what happened?
- In Simulated Annealing if the current state's evaluation function value is same as the neighbor state , how does the probability of moving to this neighbor change as the temperature, T , is gradually decreased?